

Education: MSc. Artificial intelligence and Data Science, University of Hull United Kingdom

PROFESSIONAL SUMMARY

Experienced **AI/ML Engineer, Data Engineer & Data Scientist** with 5+ years of experience delivering high-impact, production-grade **AI systems, intelligent agents, and scalable data pipelines** across **AWS, Azure, and GCP**, with proven expertise in architecting end-to-end ML solutions leveraging **LangChain, LlamaIndex, Hugging Face, AWS Bedrock, SageMaker, Vertex AI, and Azure ML Studio**. Adept at building **autonomous AI agents, RAG pipelines, and LLM-powered applications** that drive measurable business outcomes, combining deep proficiency in **Python, PySpark, Terraform, SQL, and MLOps** with strong experience in **NLP, computer vision, graph deep learning, time series forecasting, and agentic AI workflows**. A strategic, collaborative leader who bridges **data engineering, AI innovation, and business intelligence**, consistently delivering solutions that are **scalable, secure, compliant, and aligned with organizational goals**. Passionate about **Ethical AI, responsible ML governance, and continuous innovation** at the frontier of generative AI.

Core Strengths: Multi-Cloud AI Engineering | LLM Fine-Tuning & Deployment | MLOps & CI/CD | Agentic AI Systems | Big Data Architecture | NLP & Computer Vision | Stakeholder Communication

TECHNICAL SKILLS

AI, ML & Generative AI LangChain, LlamaIndex, LangGraph, AutoGen, Model Context Protocol (MCP), RAG Pipelines, LLMs (GPT-4, Claude, Gemini, Llama), Hugging Face Transformers, Fine-Tuning, Prompt Engineering, NLP, Computer Vision, Transformers (ViT), CLIP, SigLIP, Graph Deep Learning, Knowledge Graphs, Ontology, Agentic AI, AutoML, Time Series Forecasting, Demand Prediction, Anomaly Detection

Cloud Platforms & MLOps AWS (SageMaker, Bedrock, Glue, Lambda, Redshift, S3, Kinesis, EC2, IAM, KMS, CloudWatch, EventBridge, SNS, SQS, Athena, Step Functions), Azure (ML Studio, Data Lake, Synapse, AI Studio), GCP (Vertex AI, BigQuery, Dataflow, Agent Builder), Terraform, CloudFormation, CI/CD, Docker, Kubernetes, Jenkins, GitHub Actions, MLflow, Airflow, LocalStack

Programming & Frameworks Python, PySpark, Scala, SQL, R, Go, Java, Bash, PowerShell, YAML, Flask, Django, React, RESTful APIs, Spring Boot 3, Spring Security, React 18, TypeScript, React Query.

Data Engineering & Analytics ETL/ELT Pipelines, Data Lakes, Data Warehouses, Snowflake, Databricks, Kafka, Hadoop, Hive, Spark, MongoDB, DynamoDB, PostgreSQL, MySQL, Elasticsearch, Redshift, BigQuery, Power BI, Tableau, Google Data Studio **ML Libraries & Tools** TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, spaCy, NLTK, OpenCV, Jupyter, MLflow, Weights & Biases

Governance, Security & Best Practices Ethical AI, Data Ethics, IAM & KMS, Lake Formation, GDPR Compliance, Data Lineage, Model Monitoring, Unit Testing, Code Review, Agile/Scrum

WORK EXPERIENCE

Data & AI Engineer

Integrated Software Developers Group LTD

September 2024 – Present

- Architected and deployed AI/ML models and autonomous agents on AWS SageMaker and AWS Bedrock, integrating LLM-powered workflows that reduced manual reporting effort by over 40% and enhanced real-time decision-making capabilities.
- Led end-to-end migration of legacy reporting systems to AWS, embedding AI-driven analytics pipelines that improved data accessibility, cut operational costs, and unlocked intelligent, self-service insights for business stakeholders.
- Built and orchestrated agentic AI workflows using LangChain and Model Context Protocol (MCP), enabling automated data retrieval, summarization, and action execution across integrated enterprise systems.
- Automated and enhanced ETL processes using AWS Glue, Python, and SQL, incorporating LLM-based transformation agents that reduced manual workloads by 35% and improved pipeline reliability and data quality.
- Designed and implemented scalable data lakes and warehouses on AWS S3, Redshift, and Athena, supporting AI model training, vector storage, and multi-dimensional performance analytics.
- Developed RAG pipelines leveraging AWS Bedrock and OpenAI APIs to deliver contextually accurate, enterprise-grade question-answering systems grounded in internal business data.
- Implemented MLOps best practices including model versioning, performance monitoring via CloudWatch and MLflow, and automated retraining pipelines to ensure sustained model accuracy in production.
- Collaborated with cross-functional stakeholders to align AI/ML solutions with business objectives, presenting model insights and pipeline architectures to both technical and non-technical audiences.

Data Engineer

Giacom

April 2023 – June 2024

- Designed and deployed production-grade data pipelines on AWS using S3, Glue, Athena, Redshift, Lambda, EventBridge, SNS, and SQS, enabling scalable, automated ETL processing for enterprise-level workloads.
- Provisioned and managed cloud infrastructure using Terraform IaC, developing reusable modules that accelerated deployment timelines by 30% and ensured infrastructure consistency across environments.
- Engineered a reusable Python framework for AWS Glue Jobs, leveraging Apache Spark for highly scalable, distributed data transformation, reducing boilerplate code and improving team productivity.
- Automated file processing workflows by developing AWS Lambda functions triggered by S3 event notifications via Terraform, eliminating manual intervention and reducing processing latency.
- Implemented robust IAM roles, policies, and Lake Formation permissions, enforcing least-privilege access and ensuring data governance compliance across the data platform.
- Designed multi-layer data models (staging, conform, and summary layers) aligned with business reporting requirements, ensuring data consistency and traceability across analytical layers.
- Monitored and validated data flows using AWS CloudWatch and custom dashboards, proactively identifying and resolving pipeline failures before business impact.
- Delivered CI/CD deployment pipelines using GitHub Actions, automating infrastructure and code deployments into production with integrated testing and rollback capabilities.
- Developed automated unit tests for Terraform IaC, improving deployment confidence and reducing infrastructure regression errors.
- Collaborated with cross-functional teams including data analysts and business stakeholders via Trello, Microsoft Teams, and Slack, ensuring alignment on data integrity, pipeline enhancements, and reporting accuracy in Power BI and Athena.
- Designed data architecture diagrams for end-to-end data flow, facilitating clear communication between engineering and business teams.

Data Analyst

MTR GROUP LIMITED

October 2022 – March 2023

- Designed and delivered interactive Tableau dashboards to visualize partner sales activity and KPIs, enabling data-driven decisions for senior stakeholders and reducing reporting turnaround time by 25%.
- Conducted exploratory data analysis (EDA) on multiple business datasets, uncovering actionable trends and anomalies that directly informed commercial strategy.
- Championed data integrity best practices across reporting workflows, delivering consultation sessions that improved analytical accuracy and stakeholder confidence in data outputs.
- Facilitated Tableau training sessions for business users, improving self-service reporting adoption and reducing dependency on the analytics team for routine reports.
- Authored Service Level Agreements (SLAs) with source systems to formalize data usage standards, accountability, and reporting timelines.
- Produced clear data model documentation, ensuring transparency and reproducibility across all analytical processes.

Python Developer

Integrated Software Developers Group LTD

January 2020 – August 2021

- Designed and developed scalable Python-based applications using Django and Flask, delivering robust web solutions on time and within scope following Agile/Scrum methodologies.
- Reduced application processing time by 30% through algorithmic optimization and refactored Python code, directly improving system throughput and user experience.
- Decreased software defect rate by 20% by implementing automated testing frameworks, elevating overall product quality and release confidence.
- Integrated RESTful APIs from external data sources, enabling seamless third-party data ingestion and enrichment within business applications.
- Managed database design, optimization, and administration across SQL (PostgreSQL, MySQL) and NoSQL (MongoDB) databases, ensuring high performance and data integrity.
- Conducted structured code reviews via GitHub PRs, mentoring junior developers and enforcing adherence to clean code standards and best practices.
- Participated in the full SDLC from requirements gathering and system design through to deployment, monitoring, and maintenance ensuring delivery alignment with business goals.
- Produced comprehensive technical documentation, facilitating knowledge transfer and reducing onboarding time for new team members.

KEY PROJECTS

OSINT Threat Intelligence Platform | University of West London & B4 Secure LTD

Tools: Python 3.11, FastAPI, LangGraph, Groq API, spaCy, ChromaDB, React 18, TypeScript, PostgreSQL, Docker, GitHub Actions, Render, Netlify

- Built a full-stack AI-driven OSINT platform for automated cyber threat monitoring and risk assessment, delivered as an open-source project. Deployed backend on Render via Docker with PostgreSQL persistence and frontend on Netlify with GitHub Actions CI/CD pipeline. Implemented a LangGraph multi-agent orchestration system, a RAG-powered AI Copilot via Groq (llama-3.1-8b-instant) and ChromaDB for context-aware threat intelligence queries. Ingests live threat data from CISA/NVD, RSS feeds (BleepingComputer, SecurityWeek) and NewsAPI, with spaCy NLP for IOC extraction, entity recognition and MITRE ATT&CK mapping across a real-time analyst dashboard.
 - [Live App](#)
 - [GitHub Repository](#)
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Clinical Trial Platform | Barts Cancer Institute, Queen Mary University of London.

Tools: Java 21, Spring Boot 3, Spring Security, React 18, TypeScript, React Query, H2, Docker, GitHub, Render, Netlify

- Built a full-stack clinical research platform for managing clinical research studies and patient recruitment, delivered as an open-source project for the Barts Cancer Institute at Queen Mary University of London. Deployed backend on Render via multi-stage Docker build and frontend on Netlify with GitHub Actions CI pipeline.
 - [Live App](#)
 - [Github Repository](#)
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AI-Powered Email Assistant, MCP + Gmail API + Claude | Founders and Coders London

Tools: Python, Model Context Protocol (MCP), Claude Desktop, Gmail API, CI/CD

- Built a production-ready AI email assistant integrating Claude with Gmail via a custom MCP server, enabling intelligent email reading, drafting, and automated response generation.
 - Deployed with CI/CD pipelines ensuring continuous reliability and seamless updates.
 - [GitHub Repository](#)
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End-to-End NLP Pipeline | HM Land Registry (HMLR)

Tools: Hugging Face, spaCy, Scikit-learn, Python, YAML, CI/CD

- Built a production-grade NLP pipeline on the BBC dataset encompassing text classification, Named Entity Recognition (NER), and extractive summarization.
 - Delivered measurable outcomes including F1-scores and confusion matrix analysis, providing actionable intelligence for HMLR stakeholders.
 - Implemented YAML-driven configuration for modular, reproducible pipeline execution with CPU-optimized performance.
 - Integrated CI/CD pipelines ensuring code quality, security scanning, and scalability.
 - [GitHub Repository](#)
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AI-Powered Human Activity Recognition System | Aston University Birmingham

Tools: Python, CRF, HMM, Naive Bayes, Ensemble Models, Streamlit, CASAS Aruba Dataset

- Developed an AI-powered smart home behavior classification system using CRF, HMM, Naive Bayes Classifier, and stacked ensemble models, aligned with Cook (2010) benchmark standards.
 - Enabled real-time activity prediction, anomaly detection, and automated PDF report generation for healthcare and smart home applications.
 - [Live App](#) | [Demo Video](#)
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Handwritten Digit Recognition | CNN (MNIST)

Tools: Python, TensorFlow/Keras, Pandas, NumPy, Matplotlib

- Implemented a Convolutional Neural Network (CNN) for supervised image classification on the MNIST dataset, achieving high accuracy on handwritten digit recognition.
- Demonstrated expertise in deep learning model design, training optimization, and visual performance evaluation.
- [Kaggle Notebook](#)